

S&P 500 Regression Analysis

Multi-horizon forecast report (1m & 3m)

Data span: 1881-01-01 -> 2023-06-01 (1,710 monthly observations)

Source: Robert Shiller monthly S&P 500 dataset (github.com/datasets/s-and-p-500)

1-MONTH FORECAST

Predicted return: +1.17%

Verdict: MILDLY BULLISH (1-month) – predicts modest positive returns

Out-of-sample R^2 : +0.0022 Direction accuracy: 59.4%

3-MONTH FORECAST

Predicted return: +0.70%

Verdict: NEUTRAL (3-month) – predicts roughly flat returns

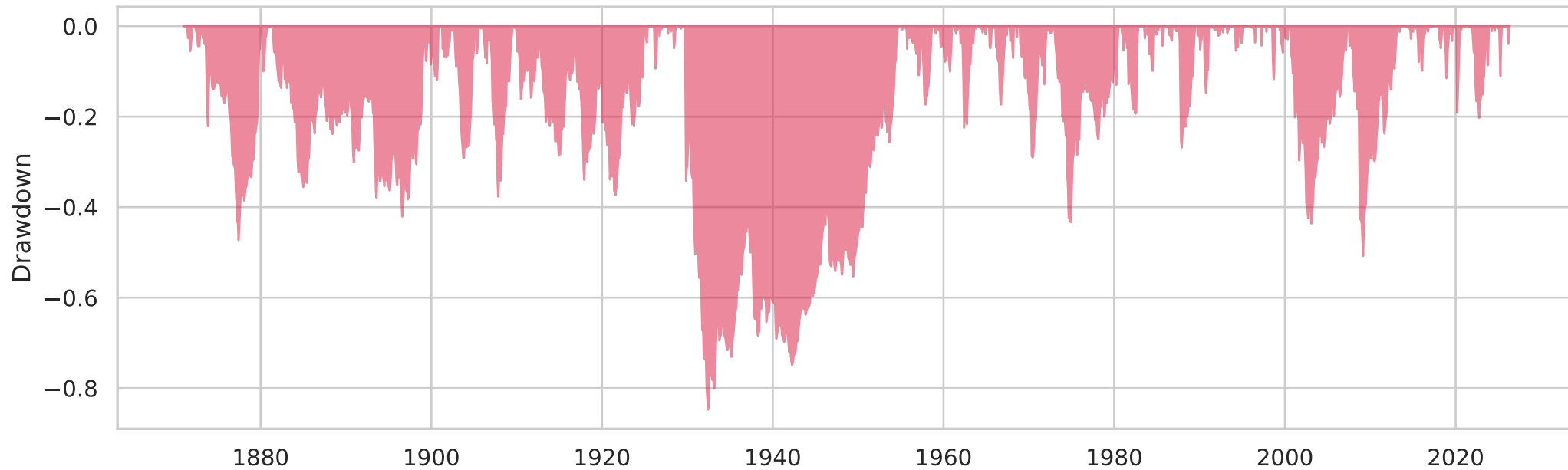
Out-of-sample R^2 : -0.1110 Direction accuracy: 51.2%

As-of feature row: 2023-06-01

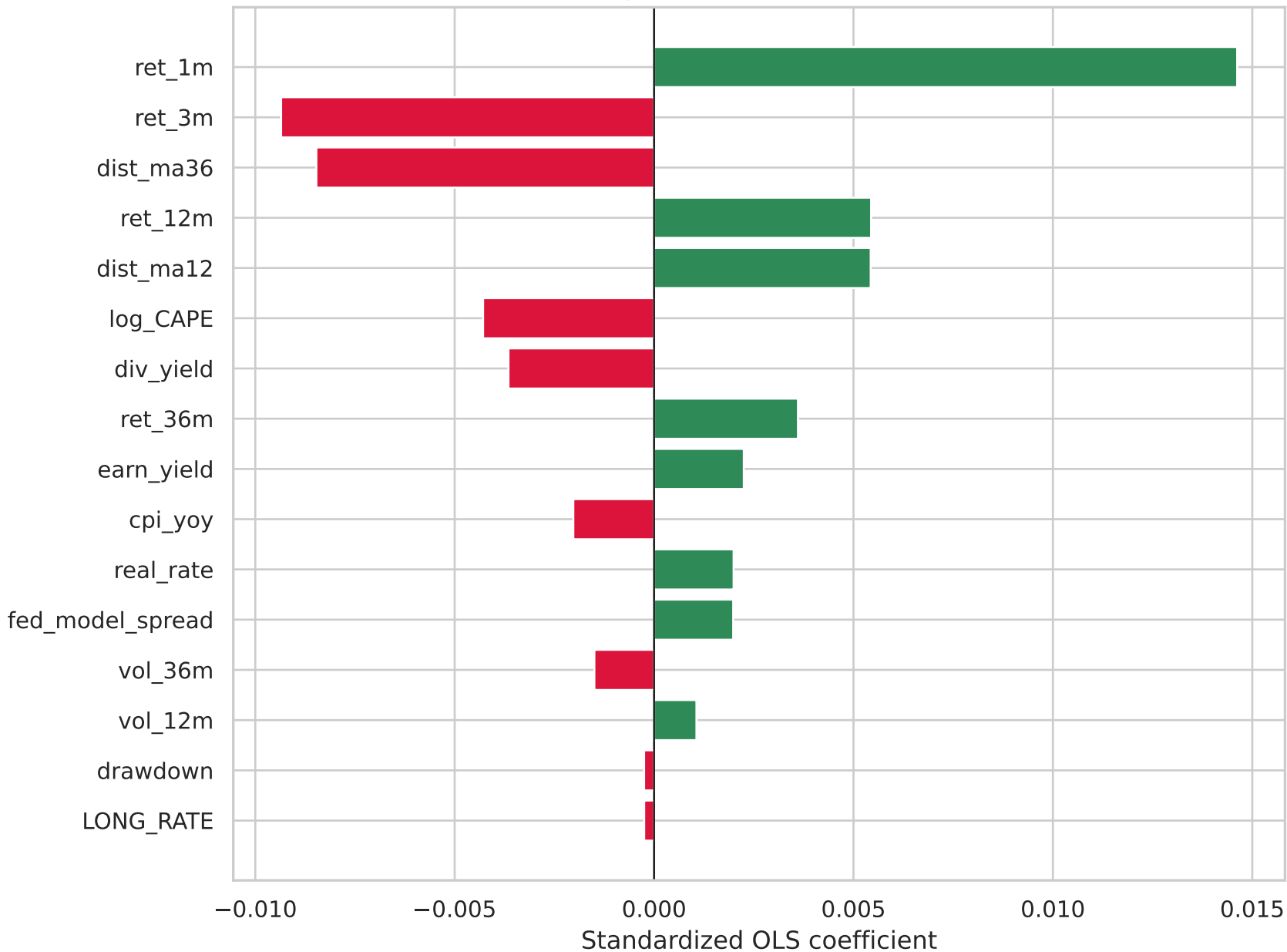
S&P 500 (nominal, log scale) -- Shiller dataset



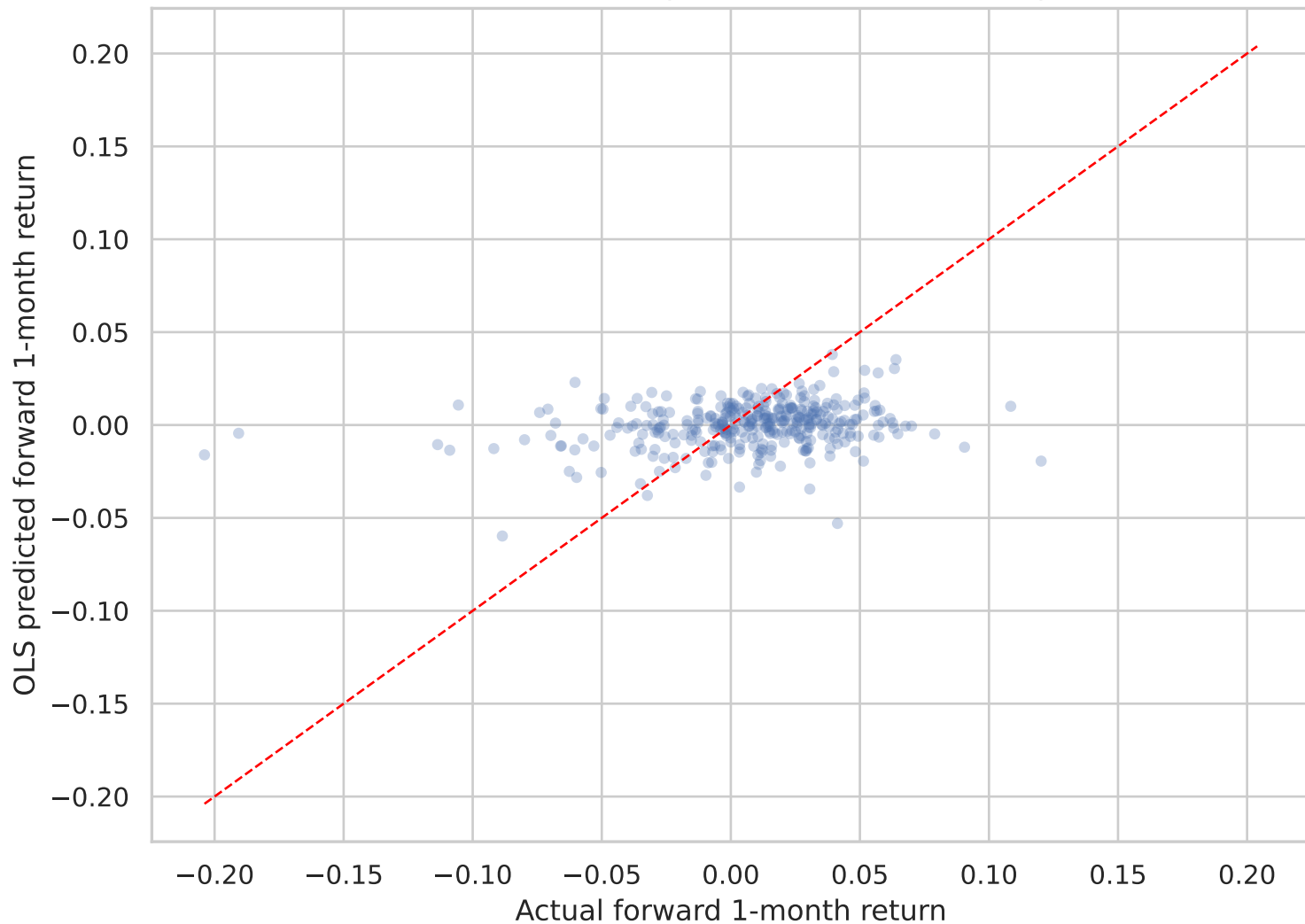
Drawdown from trailing peak



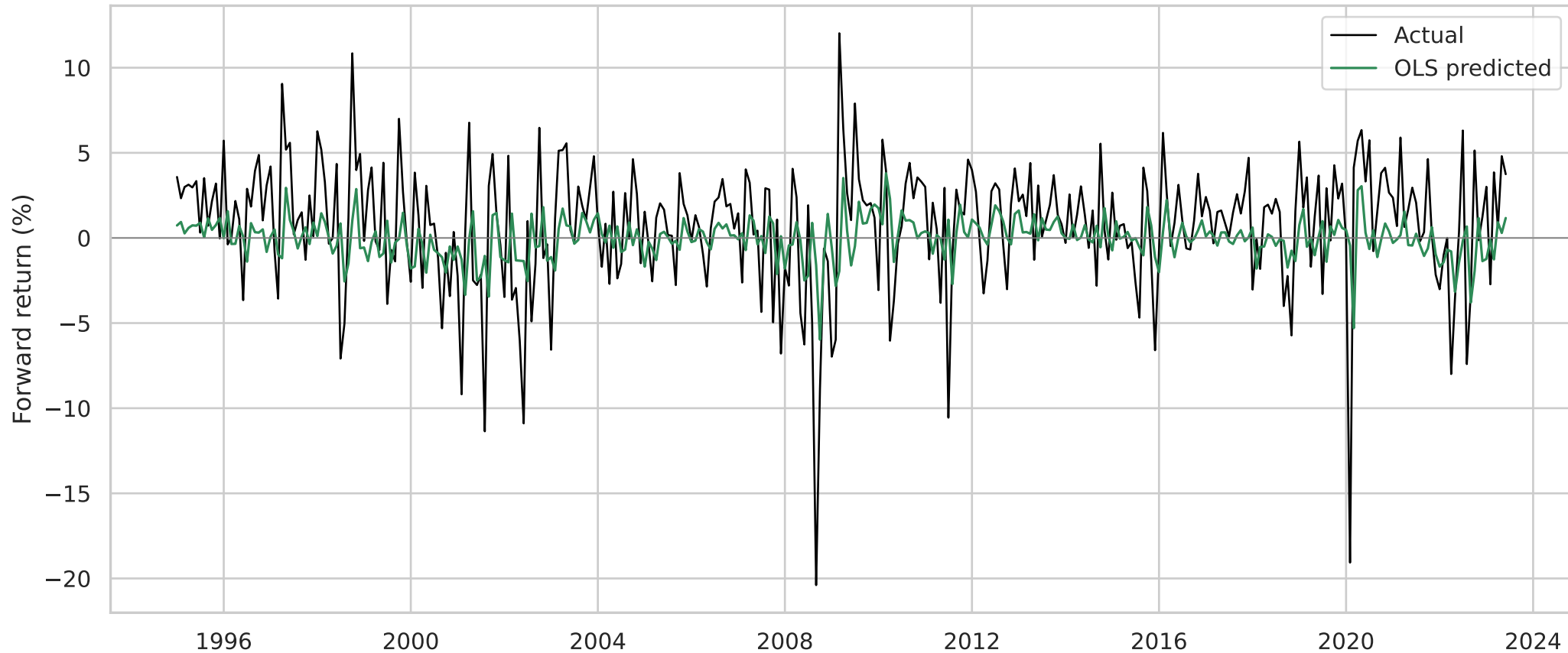
Feature impact on 1-month S&P 500 return



S&P 500 -- actual vs predicted (1m, out-of-sample)



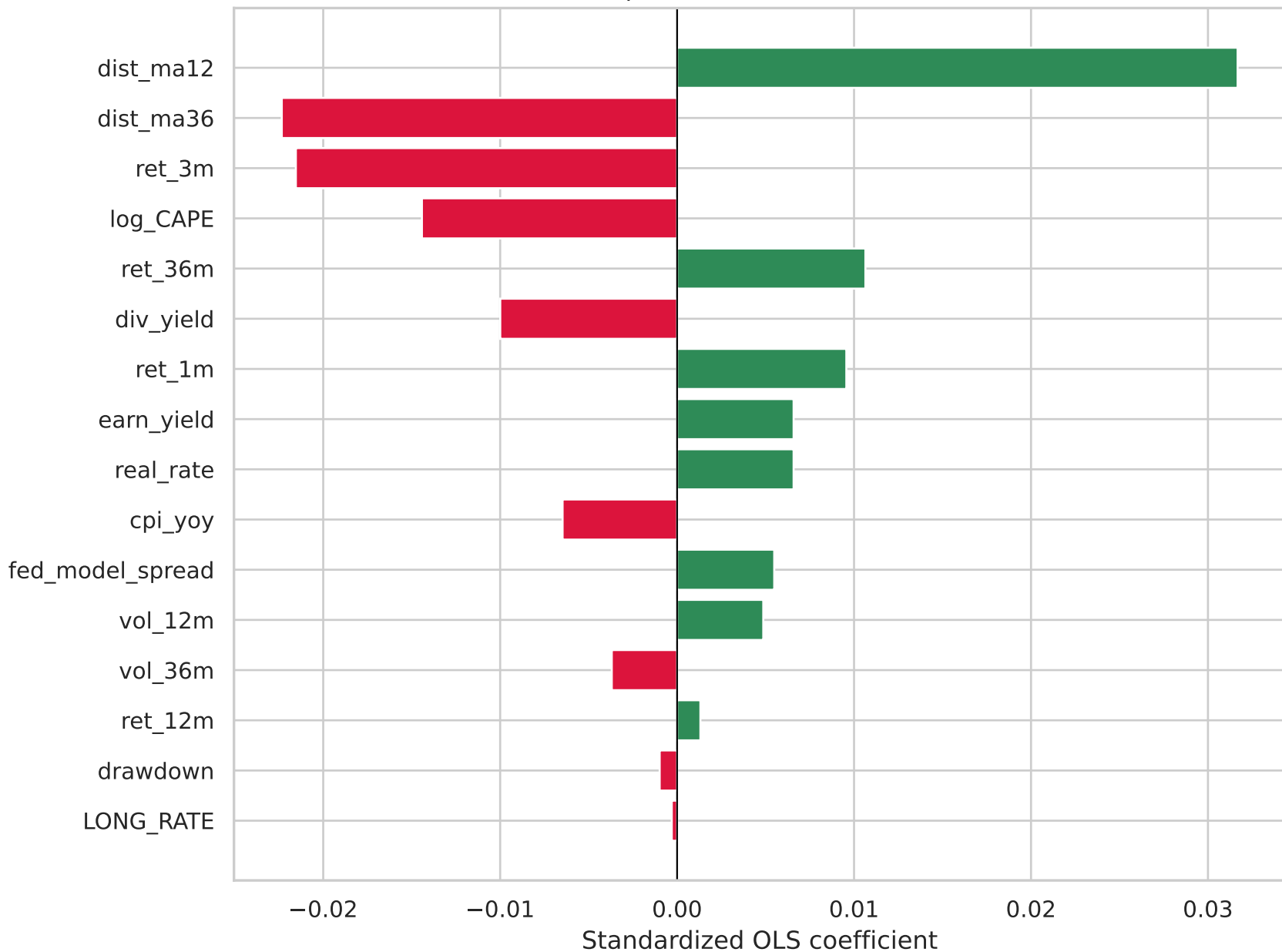
S&P 500 forward 1-month return -- actual vs predicted over time



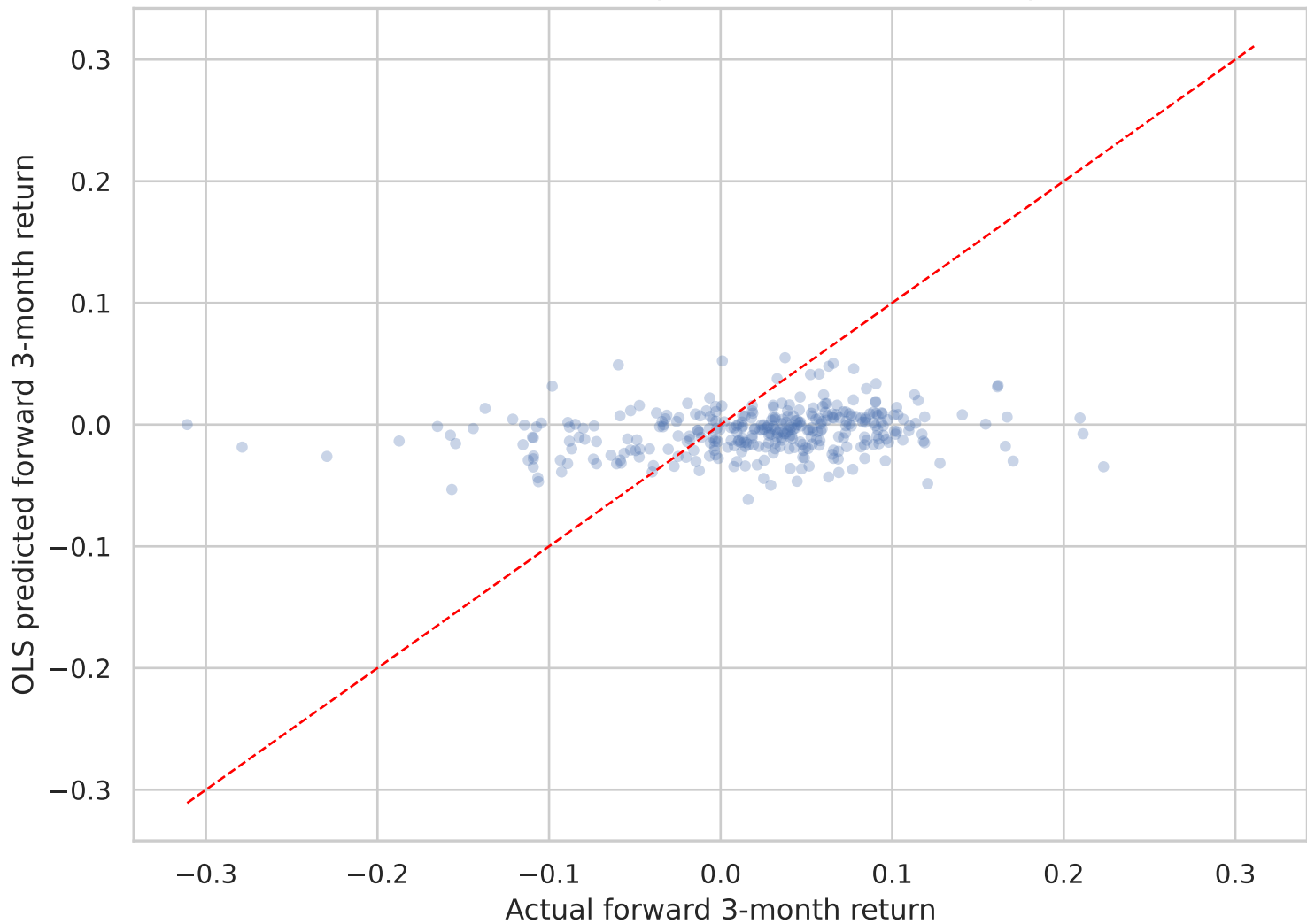
Top features -- 1-month horizon

feature	ols_standardized_coef	ols_pvalue	ridge_standardized_coef
ret_1m	+0.0146	0.0000	+0.0144
ret_3m	-0.0094	0.0000	-0.0089
dist_ma36	-0.0085	0.0191	-0.0075
ret_12m	+0.0054	0.0517	+0.0052
dist_ma12	+0.0054	0.1015	+0.0050
log_CAPE	-0.0043	0.0570	-0.0041
div_yield	-0.0037	0.1924	-0.0033
ret_36m	+0.0036	0.1067	+0.0031
earn_yield	+0.0022	0.1009	+0.0022
cpi_yoy	-0.0020	0.0070	-0.0020

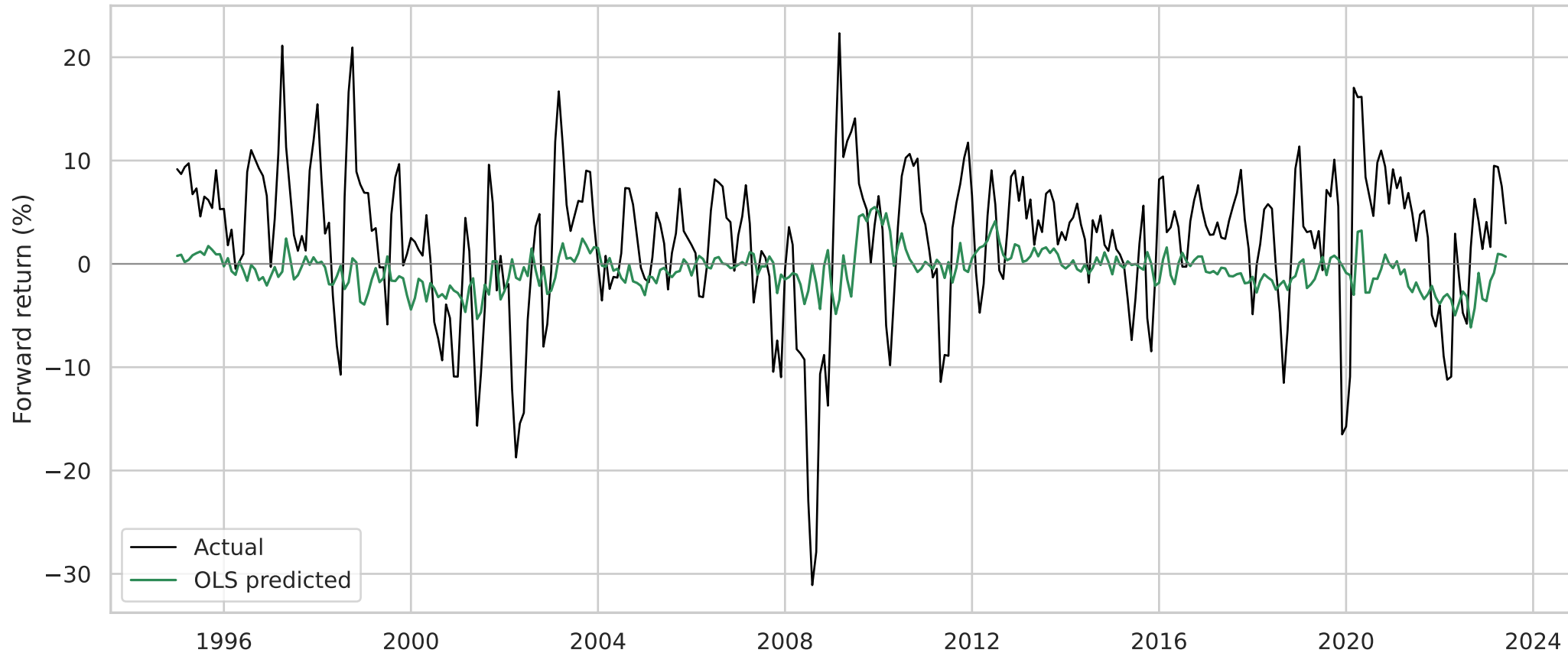
Feature impact on 3-month S&P 500 return



S&P 500 -- actual vs predicted (3m, out-of-sample)



S&P 500 forward 3-month return -- actual vs predicted over time



Top features -- 3-month horizon

feature	ols_standardized_coef	ols_pvalue	ridge_standardized_coef
dist_ma12	+0.0317	0.0000	+0.0292
dist_ma36	-0.0223	0.0030	-0.0198
ret_3m	-0.0215	0.0000	-0.0202
log_CAPE	-0.0144	0.0021	-0.0139
ret_36m	+0.0106	0.0223	+0.0093
div_yield	-0.0100	0.0866	-0.0091
ret_1m	+0.0096	0.0009	+0.0094
earn_yield	+0.0066	0.0212	+0.0064
real_rate	+0.0066	0.0000	+0.0066
cpi_yoy	-0.0065	0.0000	-0.0064

Full text summary

S&P 500 Regression Analysis -- Summary

Data: Shiller monthly S&P 500 dataset
Span: 1881-01-01 -> 2023-06-01 (1,710 monthly observations)

Forecast horizon: 1 month(s)

Train: 1881-01-01 -> 1994-12-01 (1,368 obs)
Test: 1995-01-01 -> 2023-06-01 (342 obs)
OLS R^2 (train): +0.1123
OLS R^2 (test): +0.0022
OLS MAE (test): 0.0273
OLS RMSE(test): 0.0371
Ridge R^2 (test): +0.0015
Ridge MAE (test): 0.0273
Direction-accuracy: 59.4%

Top 6 features by |standardized coefficient|:

ret_1m	std_coef=+0.0146	p=0.0000
ret_3m	std_coef=-0.0094	p=0.0000
dist_ma36	std_coef=-0.0085	p=0.0191
ret_12m	std_coef=+0.0054	p=0.0517
dist_ma12	std_coef=+0.0054	p=0.1015
log_CAPE	std_coef=-0.0043	p=0.0570

As-of: 2023-06-01
Predicted 1-month return: +1.17%
Verdict: MILDLY BULLISH (1-month) – predicts modest positive returns

Forecast horizon: 3 month(s)

Train: 1881-01-01 -> 1994-12-01 (1,368 obs)
Test: 1995-01-01 -> 2023-06-01 (342 obs)
OLS R^2 (train): +0.0760
OLS R^2 (test): -0.1110
OLS MAE (test): 0.0602
OLS RMSE(test): 0.0755
Ridge R^2 (test): -0.1132
Ridge MAE (test): 0.0604
Direction-accuracy: 51.2%

Top 6 features by |standardized coefficient|:

dist_ma12	std_coef=+0.0317	p=0.0000
dist_ma36	std_coef=-0.0223	p=0.0030
ret_3m	std_coef=-0.0215	p=0.0000
log_CAPE	std_coef=-0.0144	p=0.0021
ret_36m	std_coef=+0.0106	p=0.0223
div_yield	std_coef=-0.0100	p=0.0866

As-of: 2023-06-01
Predicted 3-month return: +0.70%
Verdict: NEUTRAL (3-month) – predicts roughly flat returns

Caveats

- Equity returns are noisy; expect single-digit R^2 even with well-known long-run predictors (CAPE, yield spreads).
- The 3-month model generally produces stronger valuation signals, because mean-reversion in CAPE/yields plays out over quarters.
- All major historical crashes (1929, 1987, 2000, 2008, 2020) are inside the dataset, but the model cannot predict the TIMING of sudden shocks -- only the expected drift given today's regime.
- Features are highly collinear; individual coefficients/p-values are interpretive guides, not causal claims.